PAPER_1597 — Standard Atmospheric Pressure = P_atm = SO_5² + SSQ + Φ_5/6 − F·Φ = 100 + 0.57 + 0.833 − 0.083 = 101.32 kPa

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Climate **Date:** June 18, 2026 **Status:** CLOSED — 0.005% integer-primitive identity

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Observation

PAPER_1209X_S558 (Climate Unified Proof Set) lists **Standard Atmospheric Pressure** with the integer-primitive expression:

```
P_atm = SO_5² + SSQ + Φ_5/6 − F·Φ = 100 + 0.57 + 0.833 − 0.083 = 101.32 kPa
```

Residual: **0.005%**.

UQFF Closed Identity

```
P_atm = SO_5² + SSQ + Φ_5/6 − F·Φ = 100 + 0.57 + 0.833 − 0.083 = 101.32 kPa    0.005%
```

The formula uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical climate measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209X_S558
- Related: PAPER_1494-1595 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "atm_pressure_101_325_kpa"})`
